Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

Team Members:

Perciyal P

Mercy A

Mukila R

Nandini H

Script-Controlled ACL – Restrict Record Access Based on Field Value

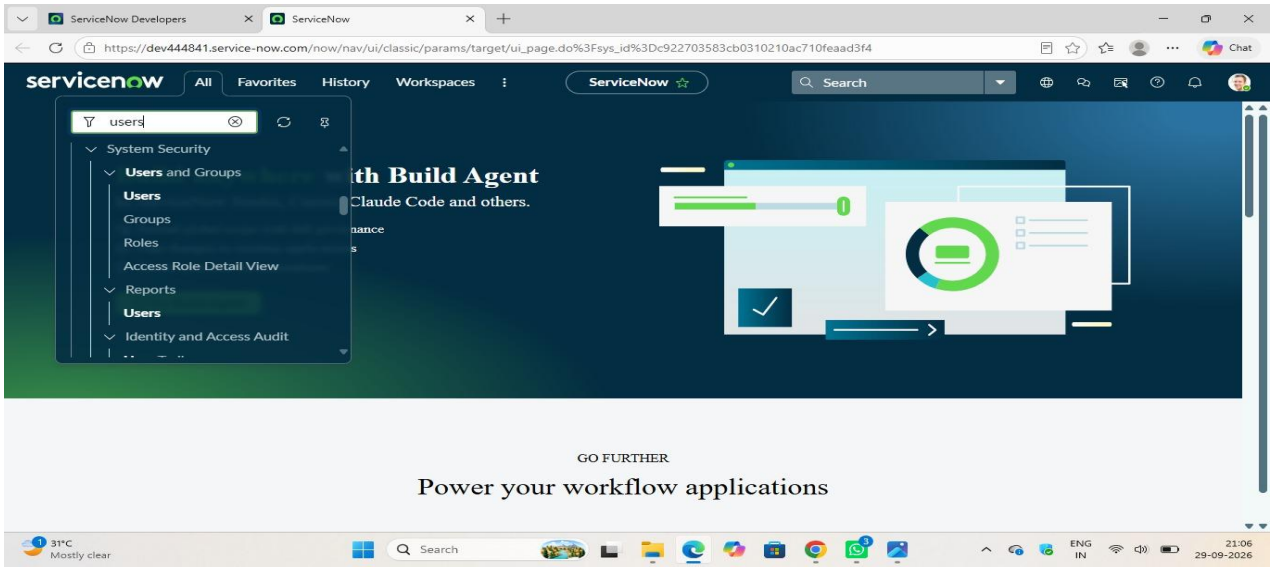
DISCRIPTION:

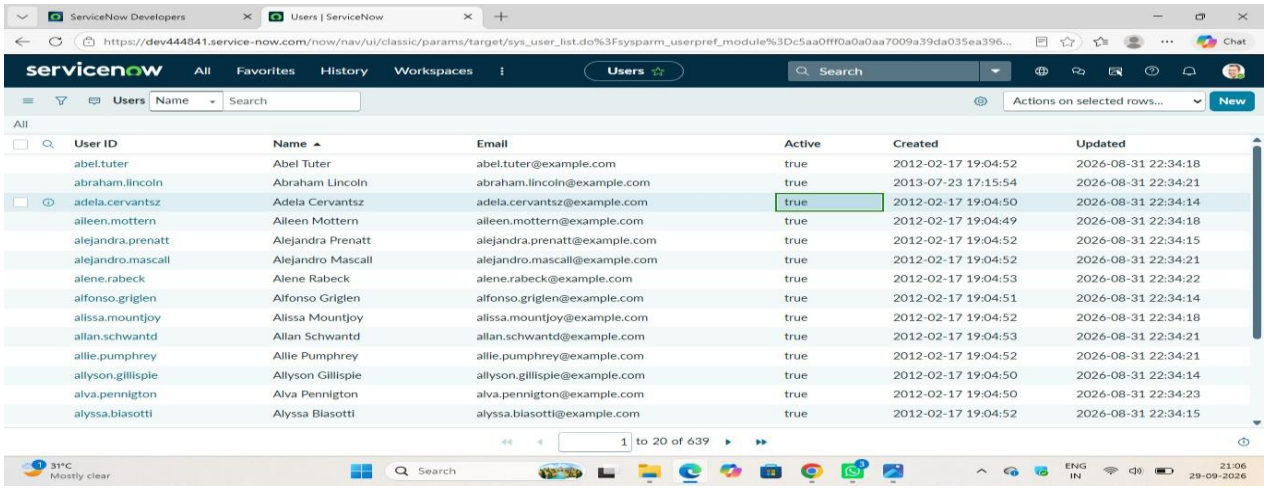
This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only Users belonging to the **EEE branch** are allowed to view records where the **Branch field is set to EEE**, while administrators retain full access.

MILESTONE – 1:CREATION OF USERS AND ROLES

USER CREATION:

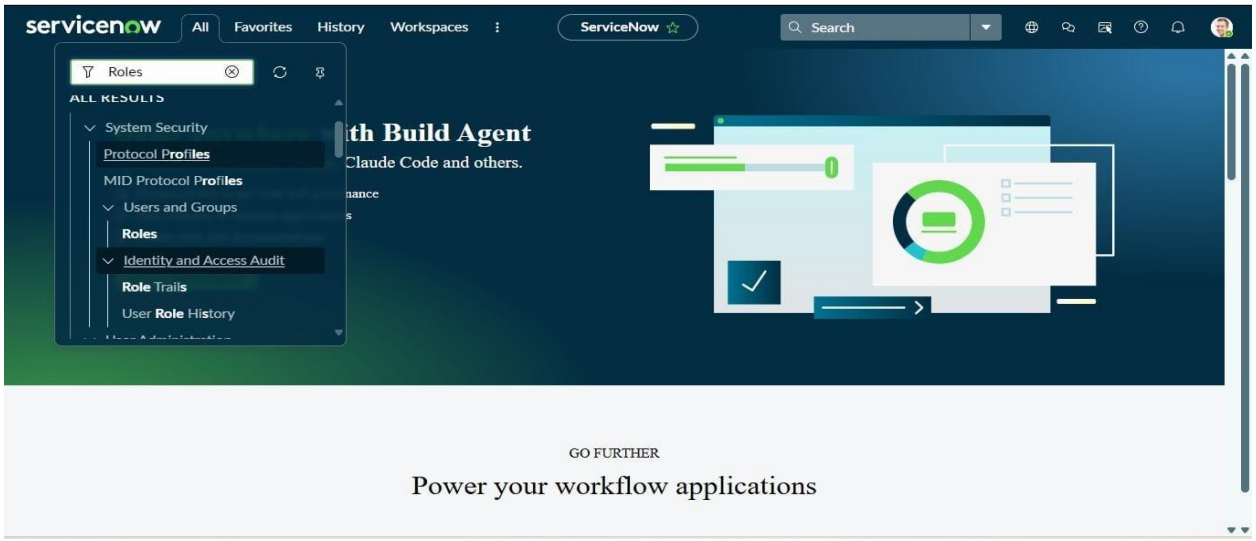
1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**
    - Navigate to **User Administration → Users**
    - Click **New**
    - **User ID**: EEE User
    - **First name**: EEE
    - **Last name**: User
    - **Email**: eeeuser@gmail.com
    - Save the record

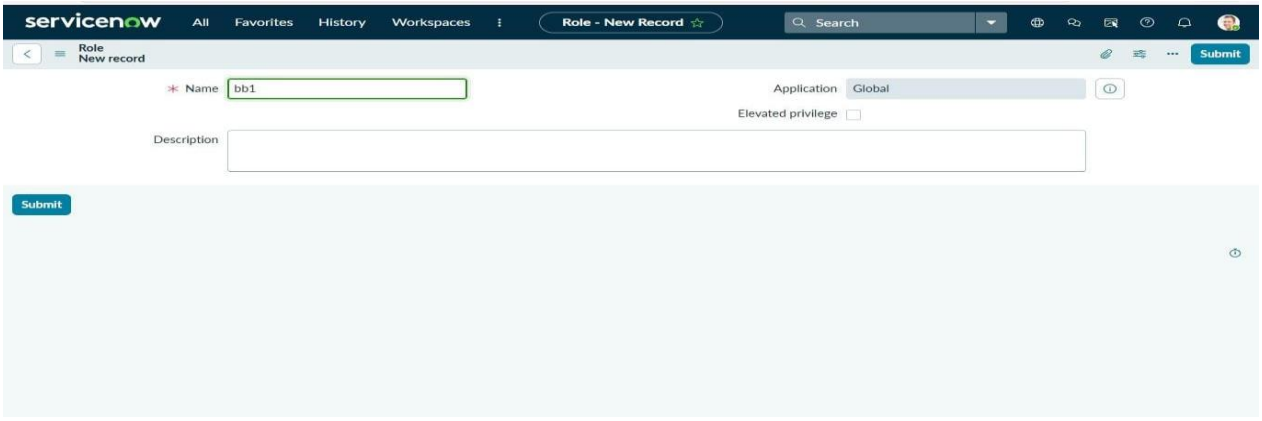




ROLES CREATION:

- Navigate to **User Administration → Roles**
- Click **New**
- **Name**: Enter role name as bb1
- **Description**: Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assigning all the roles to the EEE User

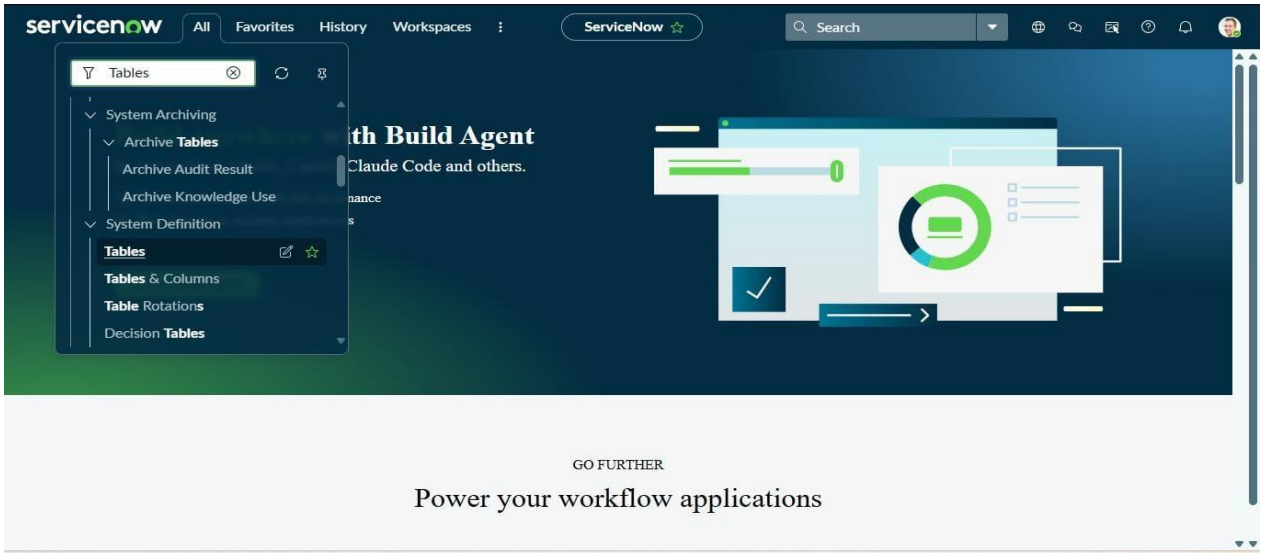


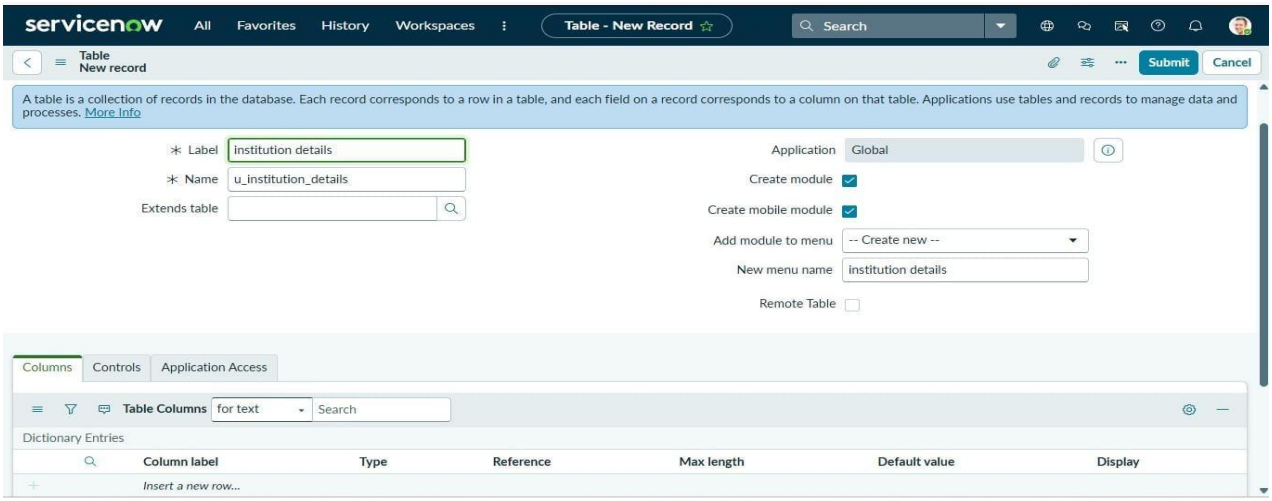


TABLES CREATION:

1. In the **Application Navigator**, search for **System Definition → Tables**.
2. Click **New** and create a custom table with the following details:

    - **Label:** Institution Details
    - **Name:** u_institution_details
    - **Extends for:** false

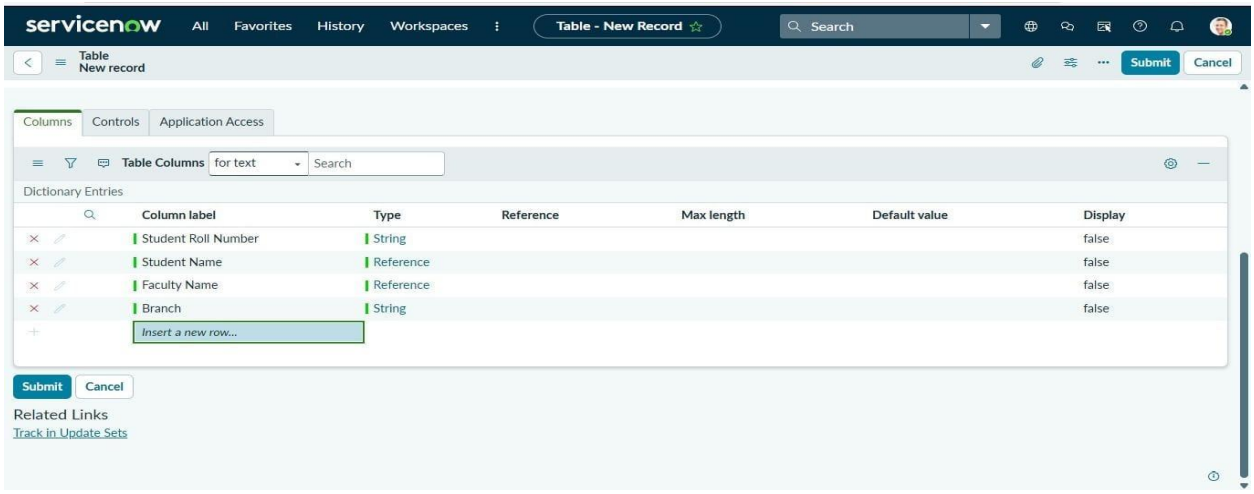




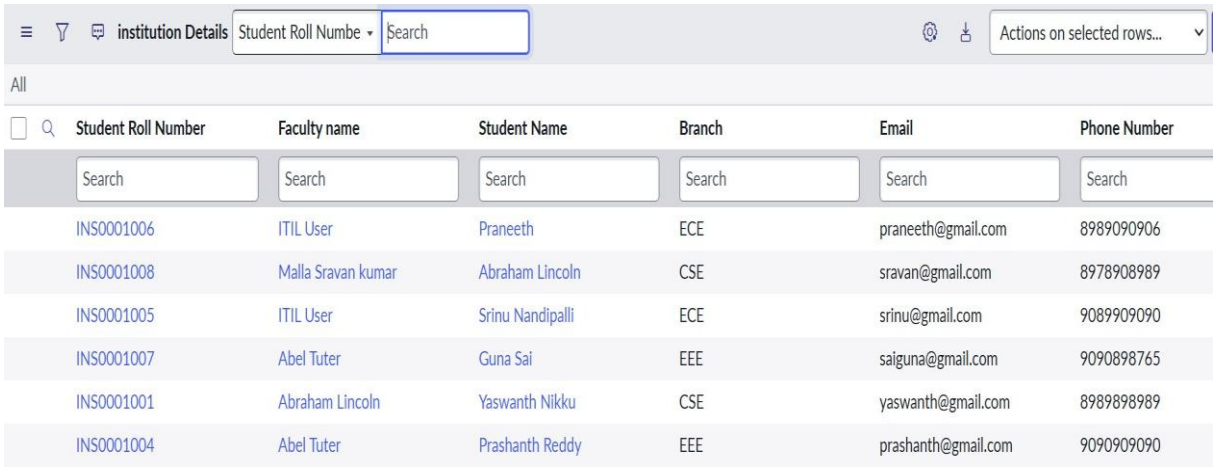
3. Click **Submit**.
4. Add the following fields to the **u_institution_details** table:

    - **Student Roll Number**– Auto Number
    - **Student Name** – Reference- User
    - **Faculty Name** – Reference- User
    - **Branch** – Choice (ECE, EEE, CSE)
    - **Email**– String
    - **Phone Number**– String
    - **Description**– Multi String

5. Save the Table



6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).



CREATION OF ACCESS CONTROL LISTS (READ ACL)

1. In the **Application Navigator**, search for **System Security → Access Control (ACL)**.
2. Elevate “Security_admin” role
3. Click **New** to create a new ACL.
4. Enter the following details:

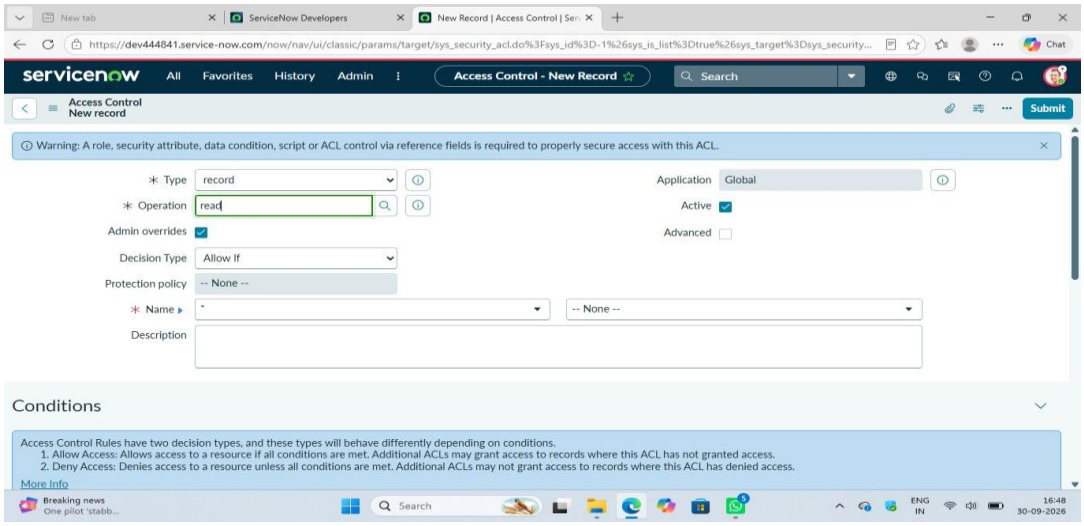
- **Type:** record
- **Operation:** read
- **Name:** u_institution_details

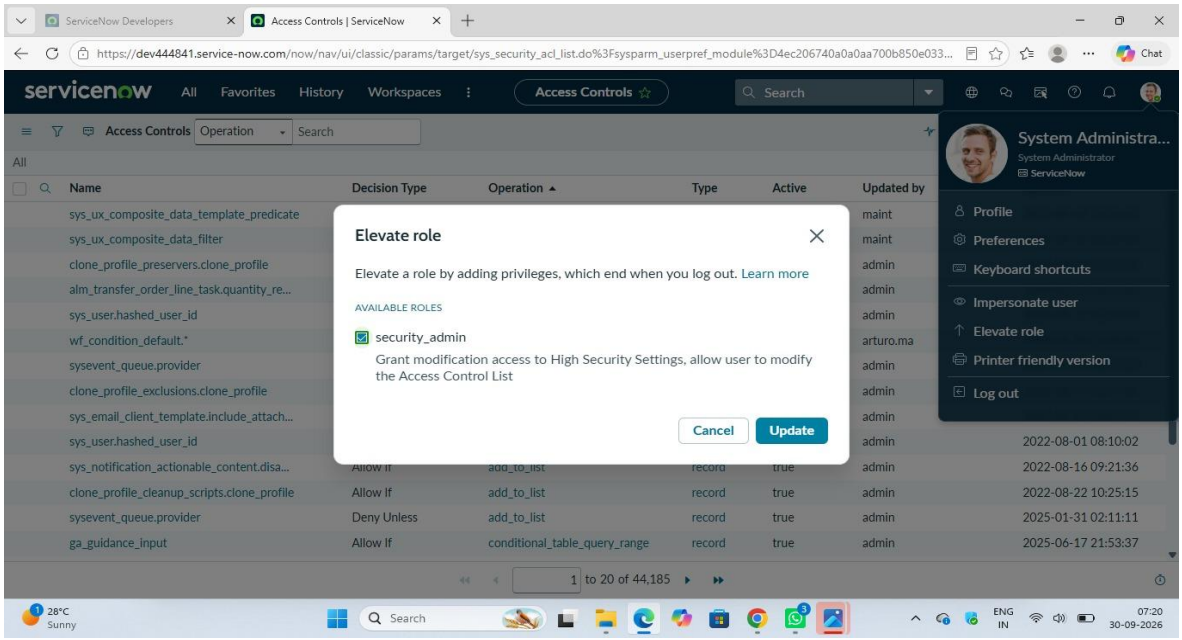
- **Active:** true
- **Advanced:** true

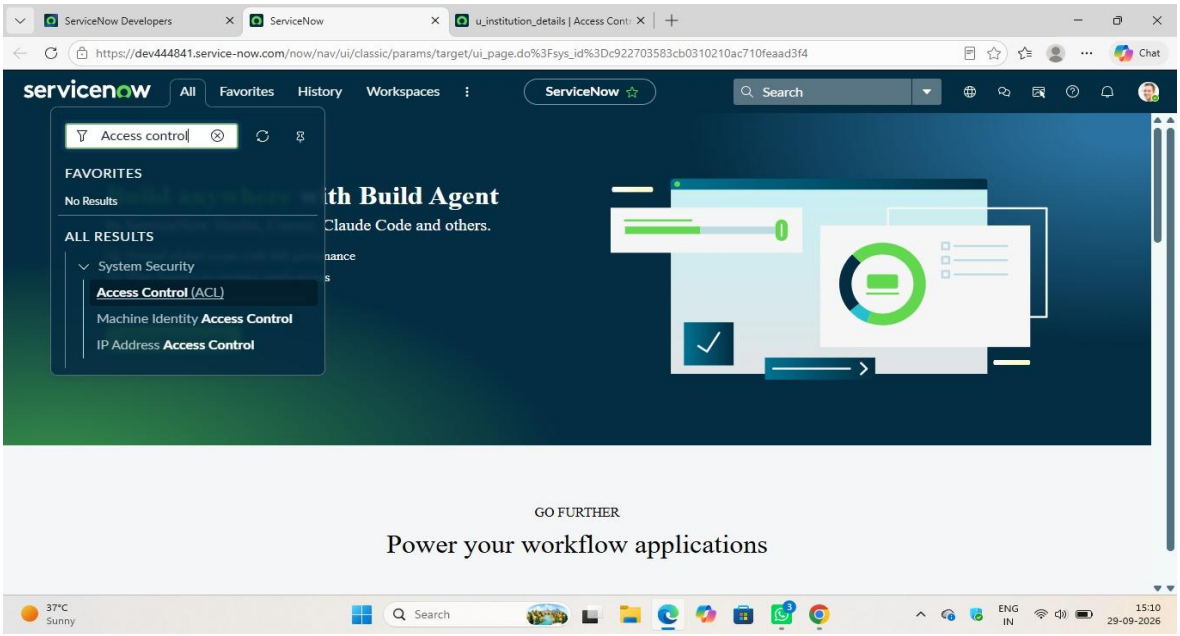
5. In the **Requires role** related list, add:
    - bb1 (custom role)
6. Add Data condition as **Branch is EEE**
7. In the **Script** field, add the following script:

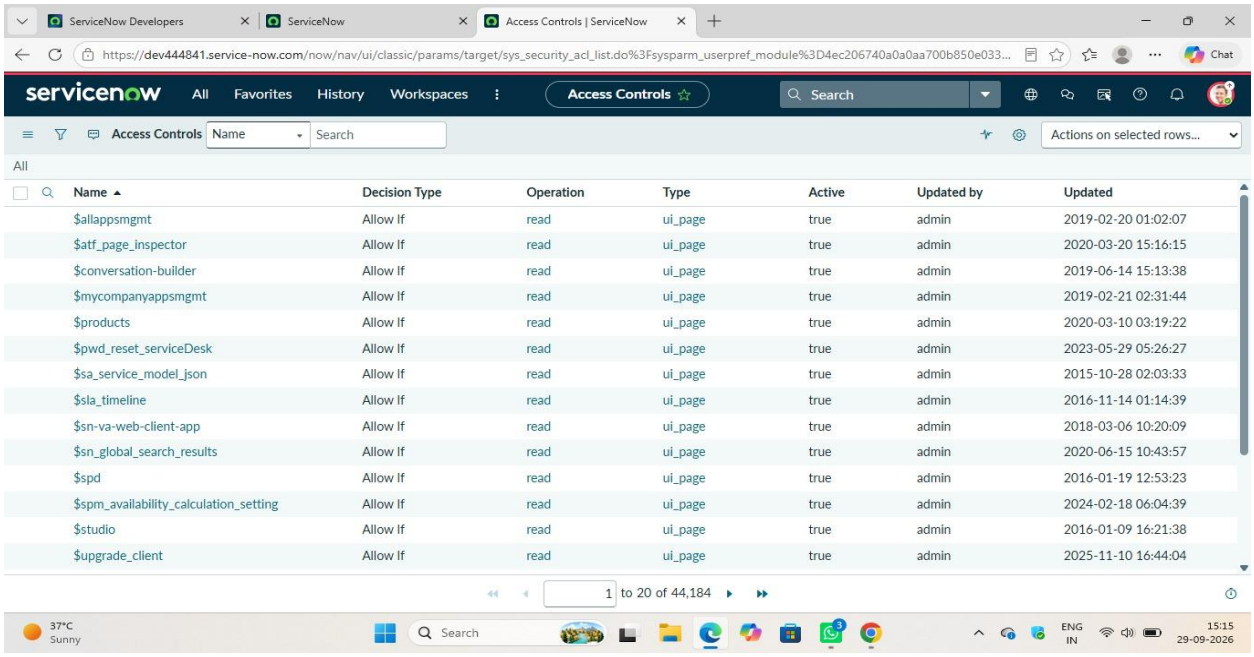
```
(function ()
 // Allow admin users full access
 if (gs.hasRole('admin')) {
    return true;
 }
 // Allow only EEE branch users to see EEE records
 if (gs.hasRole('bb1')){
    return true;
 }
 // Deny access for all others
 return false;
})();
```

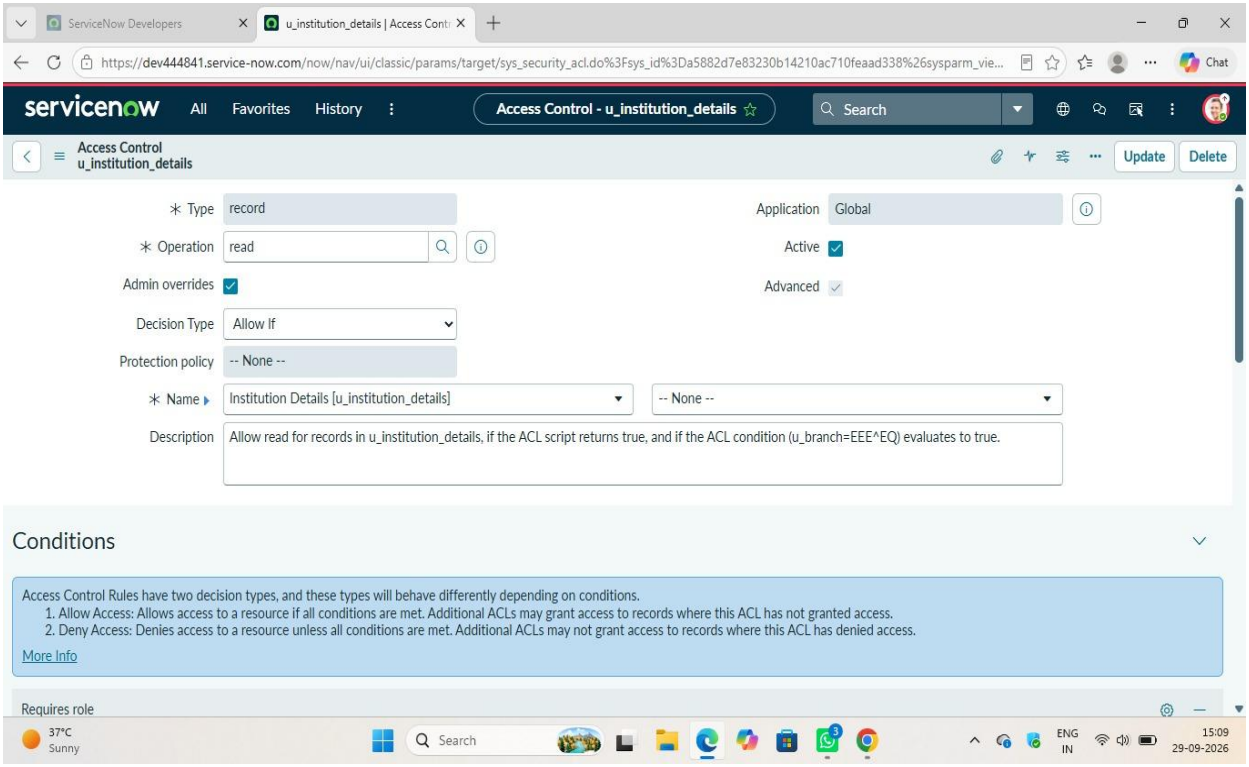
8. Click **Submit** to save the ACL.




Elevate role
Elevate a role by adding privileges, which end when you log out. Learn more
AVAILABLE ROLES
security_admin
Grant modification access to High Security Settings, allow user to modify the Access Control List
Cancel
Update


Access control
FAVORITES
No Results
ALL RESULTS
System Security
Access Control (ACL)
Machine Identity Access Control
IP Address Access Control
GO FURTHER
Power your workflow applications


ServiceNow Developers
ServiceNow
Access Controls | ServiceNow
https://dev444841.service-now.com/now/nav/ui/classic/params/target/sys_security_acl_list.do%3Fsysparm_userpref_module%3D4ec206740a0aa700b850e033...
servicenow
All
Favorites
History
Workspaces
Access Controls
Search
Access Controls
Name
Search
Actions on selected rows...
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
$allappsmgmt Allow If read ui_page true admin 2019-02-20 01:02:07
$atf_page_inspector Allow If read ui_page true admin 2020-03-20 15:16:15
$conversation-builder Allow If read ui_page true admin 2019-06-14 15:13:38
$mycompanyappsmgmt Allow If read ui_page true admin 2019-02-21 02:31:44
$products Allow If read ui_page true admin 2020-03-10 03:19:22
$pwd_reset_serviceDesk Allow If read ui_page true admin 2023-05-29 05:26:27
$sa_service_model_json Allow If read ui_page true admin 2015-10-28 02:03:33
$sla_timeline Allow If read ui_page true admin 2016-11-14 01:14:39
$sn-va-web-client-app Allow If read ui_page true admin 2018-03-06 10:20:09
$sn_global_search_results Allow If read ui_page true admin 2020-06-15 10:43:57
$spd Allow If read ui_page true admin 2016-01-19 12:53:23
$spm_availability_calculation_setting Allow If read ui_page true admin 2024-02-18 06:04:39
$studio Allow If read ui_page true admin 2016-01-09 16:21:38
$upgrade_client Allow If read ui_page true admin 2025-11-10 16:44:04
1 to 20 of 44,184
37°C Sunny
Search
ENG IN
15:15 29-09-2026


ServiceNow Developers
u_institution_details | Access Contr
https://dev444841.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3Da5882d7e83230b14210ac710feaad338%26sysparm_vie...
servicenow
All
Favorites
History
Access Control - u_institution_details
Search
Access Control
u_institution_details
Update
Delete
* Type record
Application Global
* Operation read
Active
Admin overrides
Advanced
Decision Type Allow If
Protection policy -- None --
* Name Institution Details [u_institution_details]
-- None --
Description Allow read for records in u_institution_details, if the ACL script returns true, and if the ACL condition (u_branch=EEE^EQ) evaluates to true.
Conditions
Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
More Info
Requires role
37°C Sunny
Search
ENG IN
15:09 29-09-2026

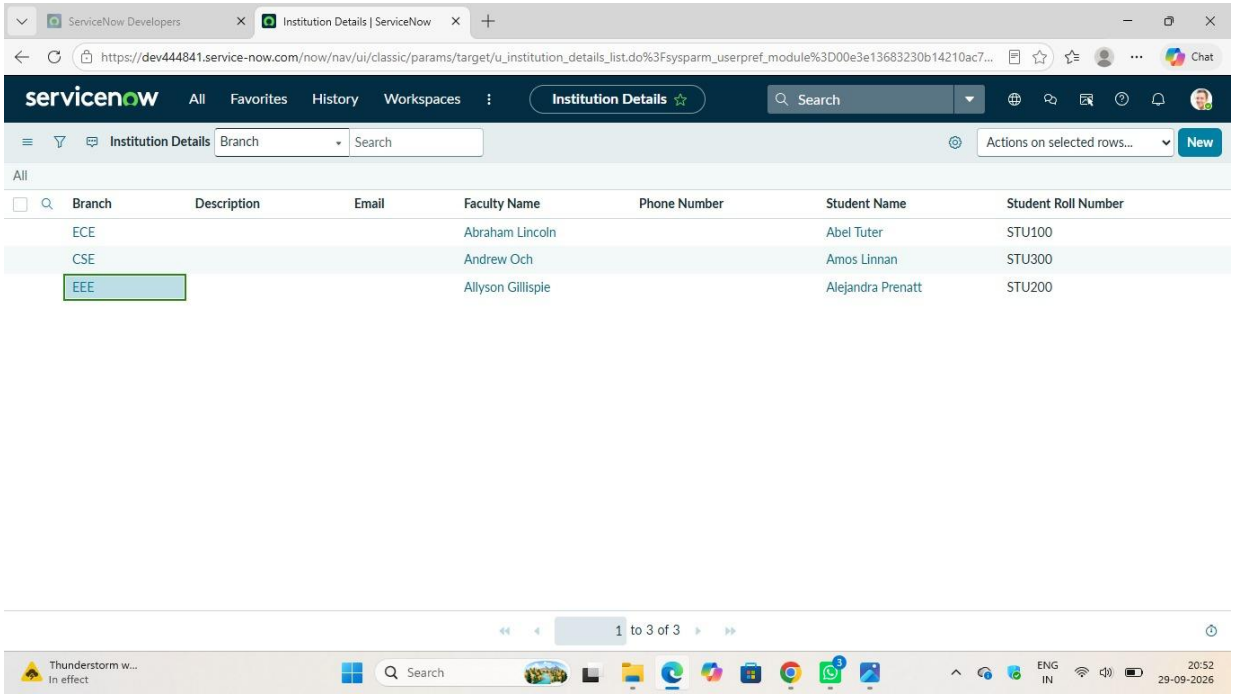
9. Impersonate **EEE users** and open the Student Records list.

10. Copy the table name u_institution_details.list/LIST and paste them all in the menu.

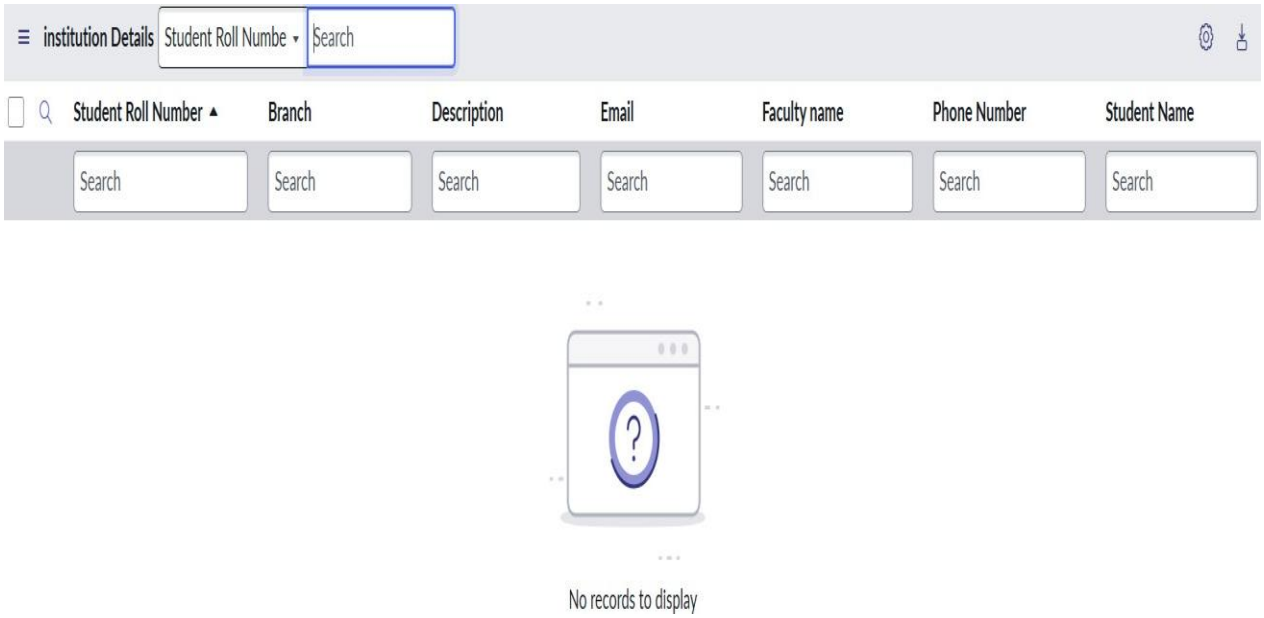
11. Impersonate **Admin** and open the Student Records list.

Verify:

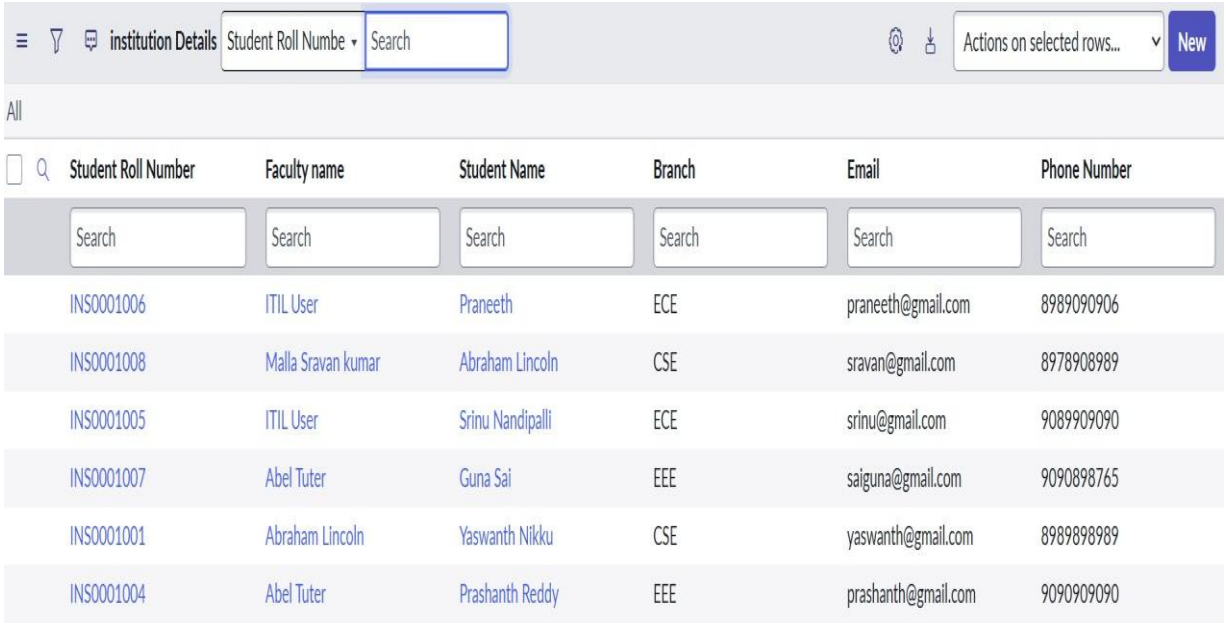
- Users with **bb1 role** can view **only EEE branch records**



- Users without the role cannot view any records

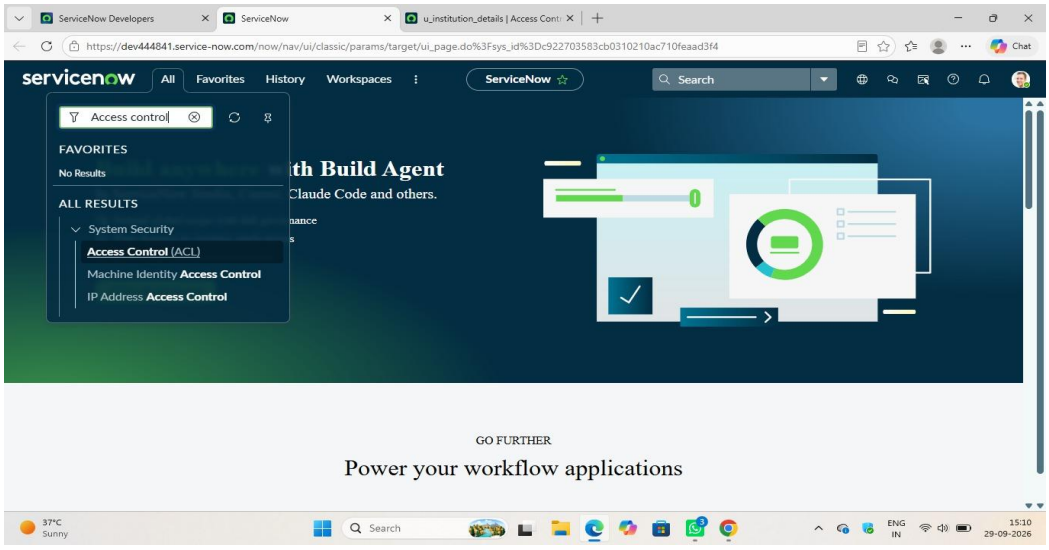


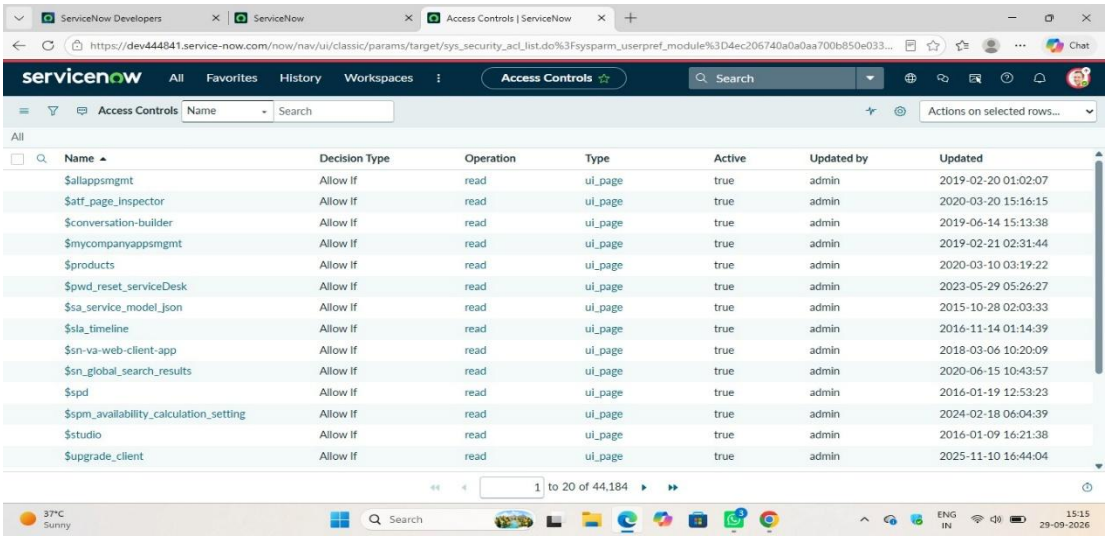
- Admin users can view **all records regardless of branch**



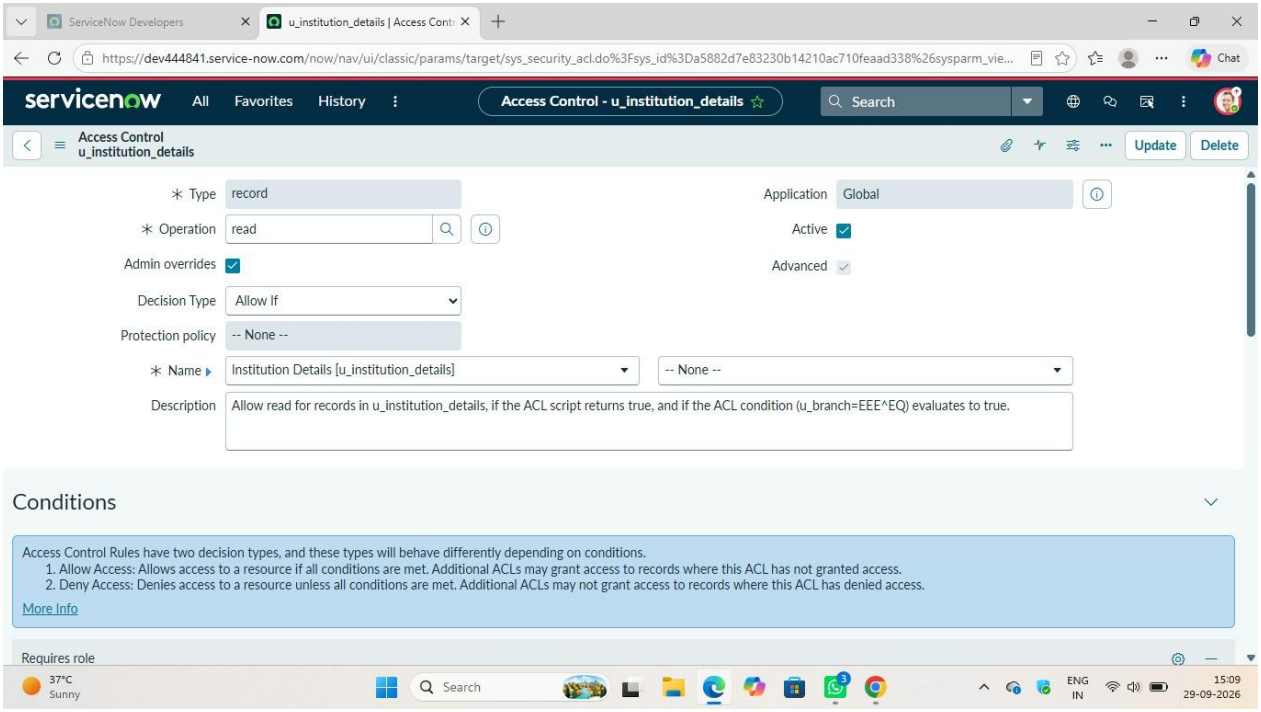
CREATION OF ACCESS CONTROL LISTS (CREATE ACL)

1. Click **New** to create a new ACL(Create)

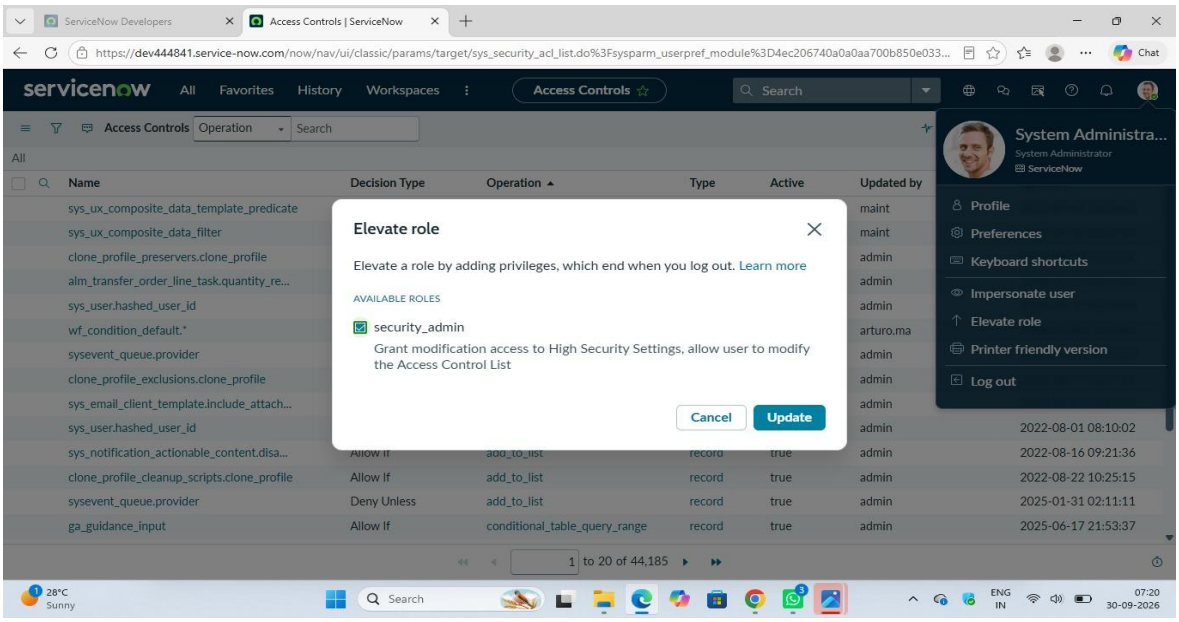




2. Enter the following details:
    - **Type:** record
    - **Operation:** create
    - **Name:** u_institution_details
    - **Active:** true
3. In the **Requires role** related list, add:
    - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



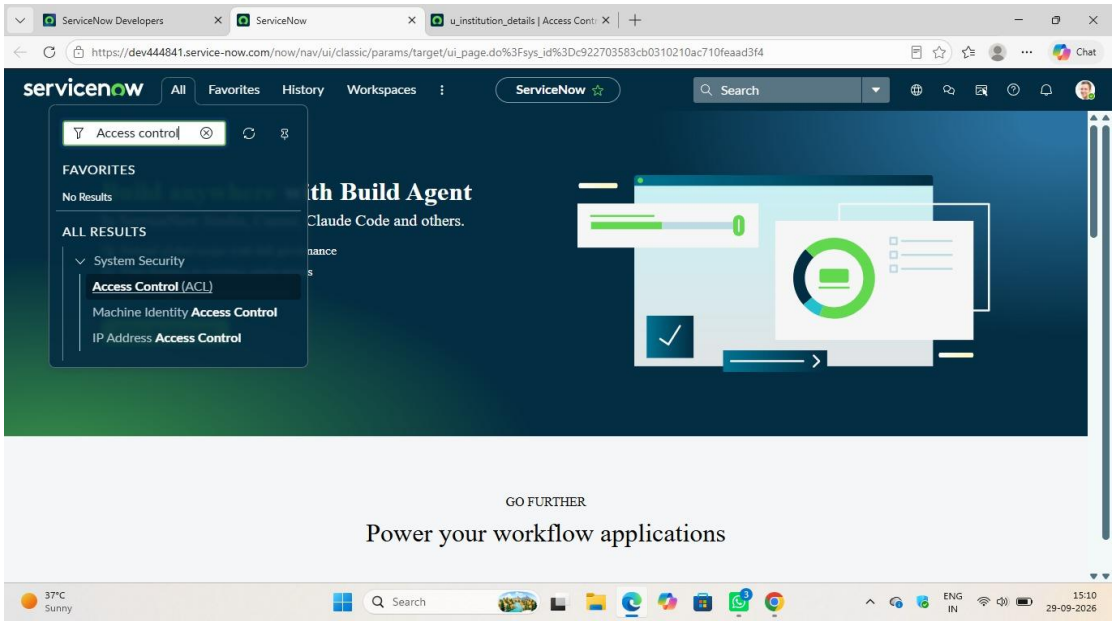
Verify:

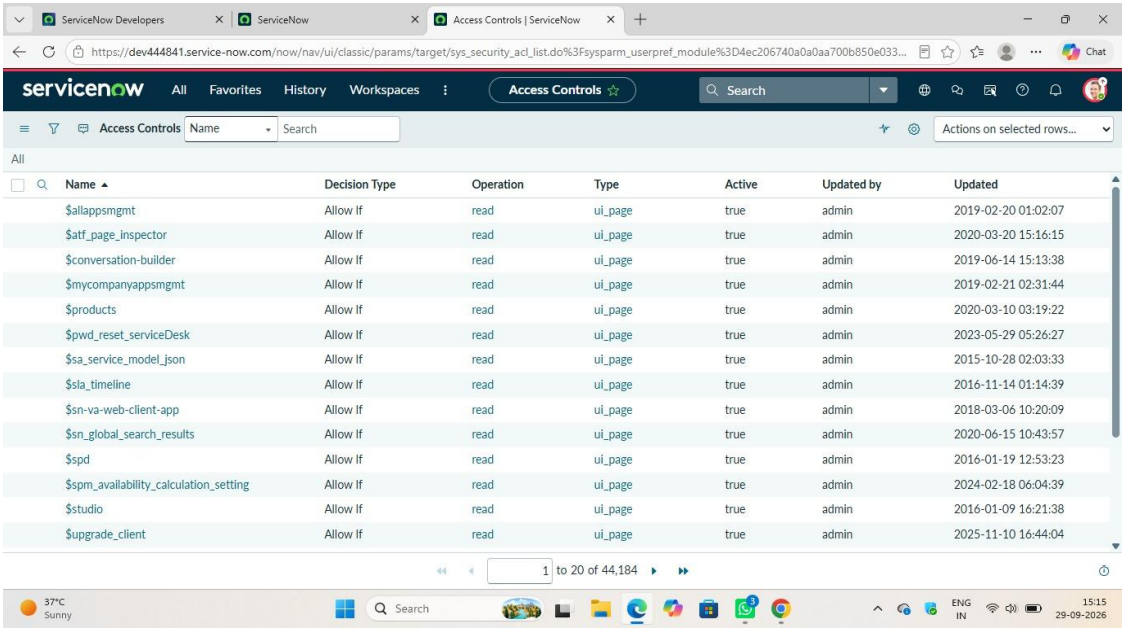


- Users with **bb1 &bb2 roles** can view **only EEE branch records b New** Button

CREATION OF ACCESS CONTROL LISTS (WRITE ACL)

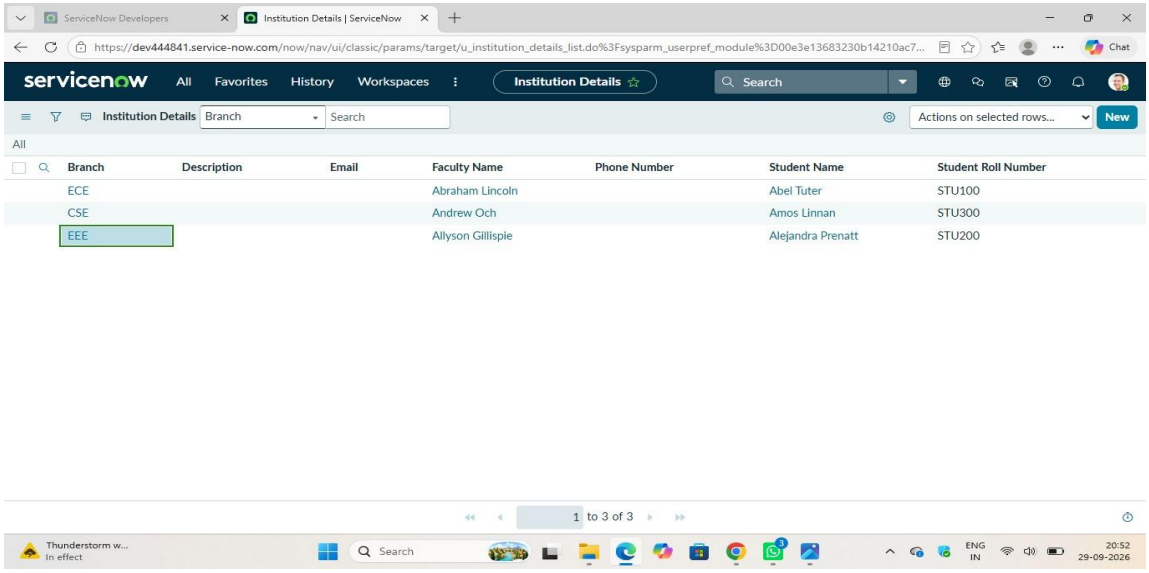
1. Click New to create a new ACL(Write)
2. Enter the following details:
    - Type: record
    - Operation: Write
    - Name: u_institution_details
    - Active: true
3. In the Requires role related list, add:
    - bb3 (custom role)
4. No need to add any Data condition
5. Save the ACL Record


ServiceNow Developers
ServiceNow
u_institution_details | Access Cont
https://dev444841.service-now.com/now/nav/ui/classic/params/target/ui_page.do%3Fsys_id%3Dc922703583cb0310210ac710feaad3f4
Chat
servicenow
All
Favorites
History
Workspaces
ServiceNow
Search
Access control
FAVORITES
No Results
ALL RESULTS
System Security
Access Control (ACL)
Machine Identity Access Control
IP Address Access Control
th Build Agent
Claude Code and others.
GO FURTHER
Power your workflow applications
37°C
Sunny
Search
ENG
IN
15:10
29-09-2026


ServiceNow Developers
ServiceNow
Access Controls | ServiceNow
https://dev444841.service-now.com/now/nav/ui/classic/params/target/sys_security_acl_list.do%3Fsysparm_userpref_module%3D4ec206740a0a0aa700b850e033...
Chat
servicenow
All
Favorites
History
Workspaces
Access Controls
Search
Access Controls
Name
Search
Actions on selected rows...
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
$allappsmgmt Allow If read ui_page true admin 2019-02-20 01:02:07
$atf_page_inspector Allow If read ui_page true admin 2020-03-20 15:16:15
$conversation-builder Allow If read ui_page true admin 2019-06-14 15:13:38
$mycompanyappsmgmt Allow If read ui_page true admin 2019-02-21 02:31:44
$products Allow If read ui_page true admin 2020-03-10 03:19:22
$pwd_reset_serviceDesk Allow If read ui_page true admin 2023-05-29 05:26:27
$sa_service_model_json Allow If read ui_page true admin 2015-10-28 02:03:33
$sla_timeline Allow If read ui_page true admin 2016-11-14 01:14:39
$sn-va-web-client-app Allow If read ui_page true admin 2018-03-06 10:20:09
$sn_global_search_results Allow If read ui_page true admin 2020-06-15 10:43:57
$spd Allow If read ui_page true admin 2016-01-19 12:53:23
$spm_availability_calculation_setting Allow If read ui_page true admin 2024-02-18 06:04:39
$studio Allow If read ui_page true admin 2016-01-09 16:21:38
$upgrade_client Allow If read ui_page true admin 2025-11-10 16:44:04
1 to 20 of 44,184
37°C
Sunny
Search
ENG
IN
15:15
29-09-2026

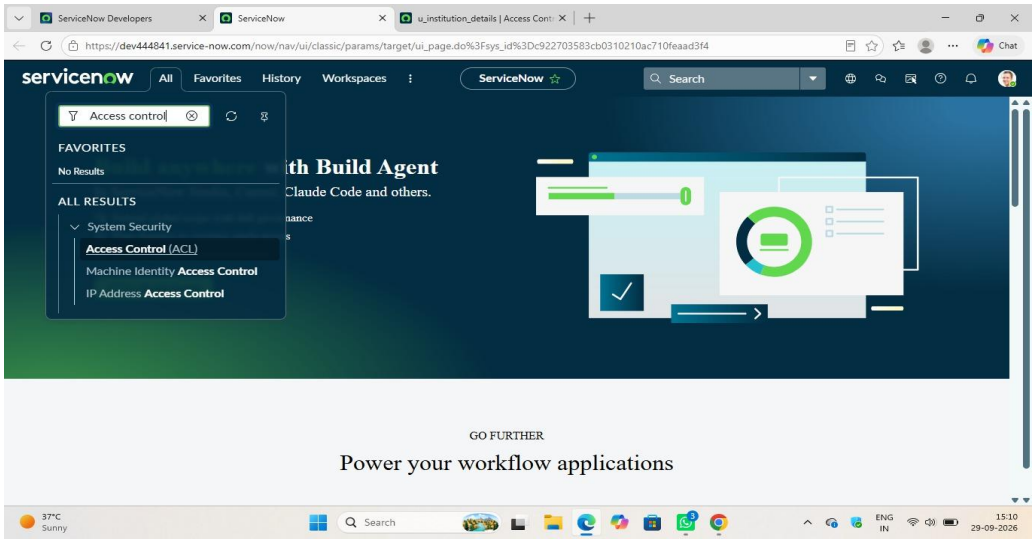
Verify:

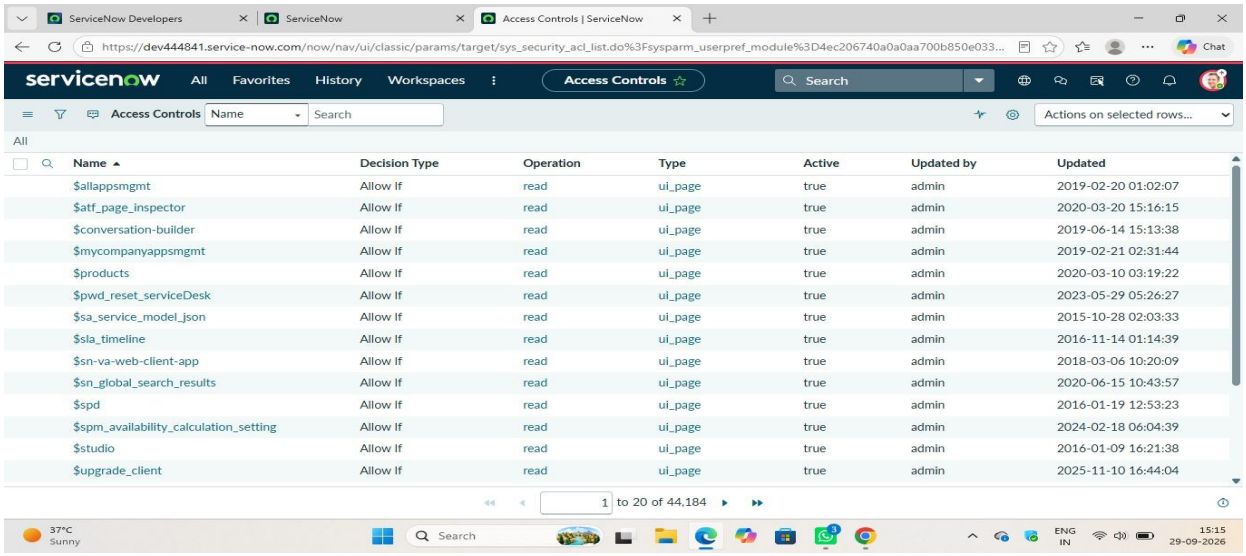
- Users with **bb1 &bb2 &bb3 roles** can view and edit the **EEE branch records along with New** Button

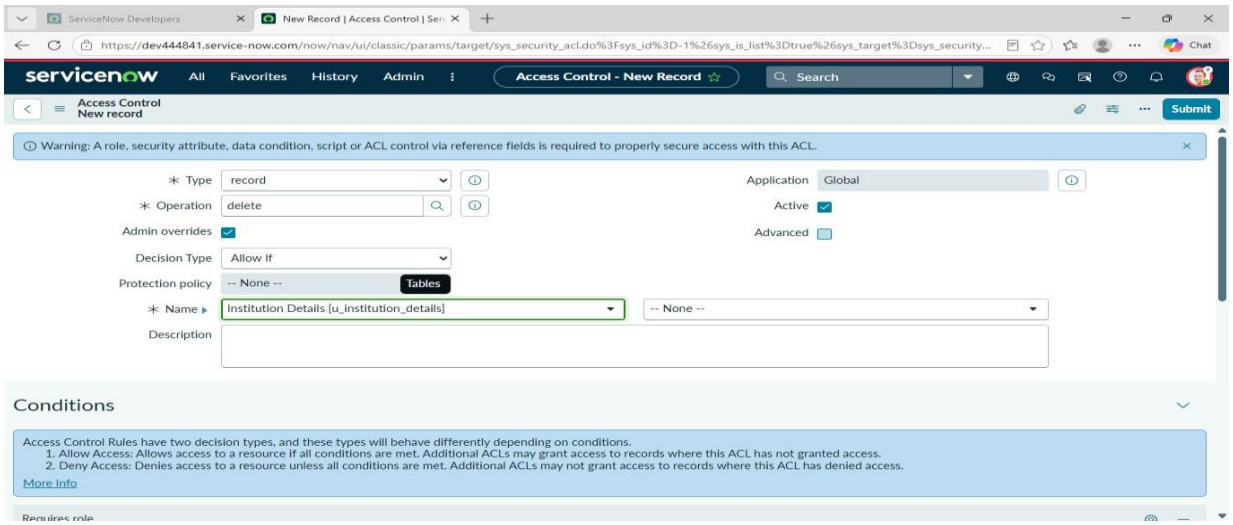


CREATION OF ACCESS CONTROL LISTS (DELETE ACL)

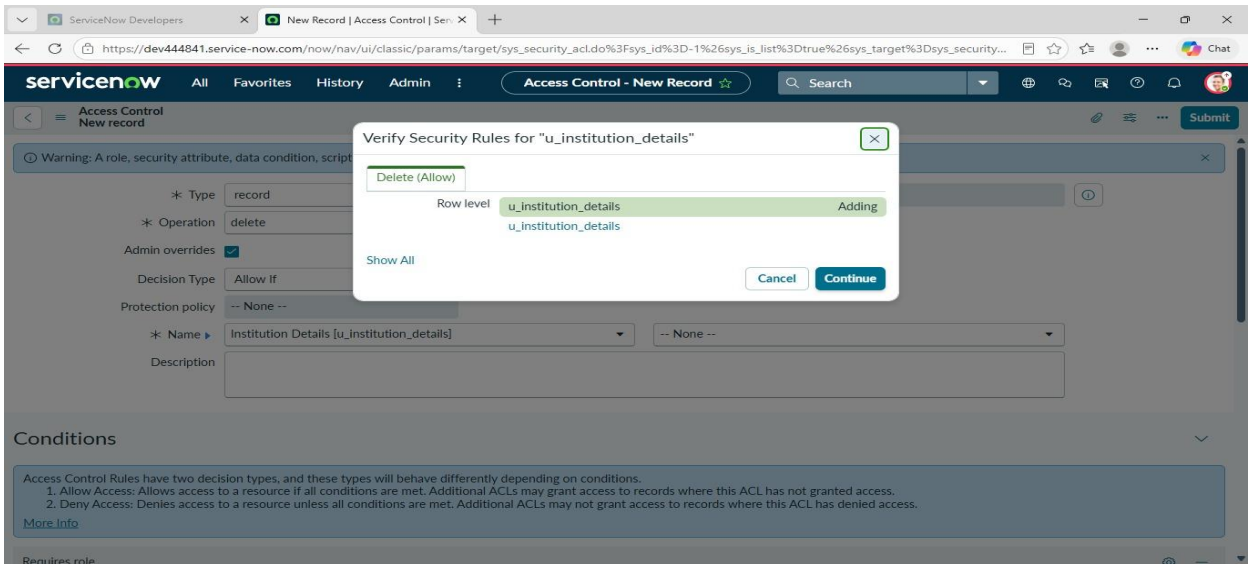
1. Click **New** to create a new ACL(Delete)
2. Enter the following details:
   - **Type:** record
   - **Operation:** Delete
   - **Name:** u_institution_details
   - **Active:** true
3. In the **Requires role** related list, add:
   - bb4 (custom role)
4. No need to add any Data condition
5. Save the ACL Record




ServiceNow Developers
ServiceNow
Access Controls | ServiceNow
servicenow
All
Favorites
History
Workspaces
Access Controls
Search
Access Controls
Name
Search
Actions on selected rows...
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
$allappsmgmt Allow If read ui_page true admin 2019-02-20 01:02:07
$atf_page_inspector Allow If read ui_page true admin 2020-03-20 15:16:15
$conversation-builder Allow If read ui_page true admin 2019-06-14 15:13:38
$mycompanyappsmgmt Allow If read ui_page true admin 2019-02-21 02:31:44
$products Allow If read ui_page true admin 2020-03-10 03:19:22
$pwd_reset_serviceDesk Allow If read ui_page true admin 2023-05-29 05:26:27
$sa_service_model_json Allow If read ui_page true admin 2015-10-28 02:03:33
$sla_timeline Allow If read ui_page true admin 2016-11-14 01:14:39
$sn-va-web-client-app Allow If read ui_page true admin 2018-03-06 10:20:09
$sn_global_search_results Allow If read ui_page true admin 2020-06-15 10:43:57
$spd Allow If read ui_page true admin 2016-01-19 12:53:23
$spm_availability_calculation_setting Allow If read ui_page true admin 2024-02-18 06:04:39
$studio Allow If read ui_page true admin 2016-01-09 16:21:38
$upgrade_client Allow If read ui_page true admin 2025-11-10 16:44:04
1 to 20 of 44,184


ServiceNow Developers
New Record | Access Control
servicenow
All
Favorites
History
Admin
Access Control - New Record
Search
Access Control
New record
Submit
Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.
Type record
Operation delete
Admin overrides
Decision Type Allow If
Protection policy -- None -- Tables
Name Institution Details [u_institution_details]
Description
Application Global
Active
Advanced
-- None --
Conditions
Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
More Info
Requires role

Verify:


ServiceNow Developers
New Record | Access Control
servicenow
All
Favorites
History
Admin
Access Control - New Record
Search
Access Control
New record
Submit
Verify Security Rules for "u_institution_details"
Delete (Allow)
Row level
u_institution_details
Adding
u_institution_details
Show All
Cancel
Continue
Type record
Operation delete
Admin overrides
Decision Type Allow If
Protection policy -- None --
Name Institution Details [u_institution_details]
Description
-- None --
Conditions
Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
More Info
Requires role

- Users with **bb1 &bb2 &bb3&bb4 roles** can see, create and edit along with delete the **EEE branch records.**

OUTCOME:

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.